

General Introduction

1. General Introduction

1.1 The ecological importance of forest microclimates in a changing climate

1.1.1 Climate change and the increasing frequency of extreme phenomena

Climate change raises mean temperature, and, more consequentially for living systems, it raises the frequency and severity of extremes. The state of the climate system and the human influence on it are now tracked and updated annually against the methods of the IPCC's Sixth Assessment. The Earth's energy imbalance has more than doubled relative to the late twentieth century (Forster et al., 2026). What that heating does to living systems is felt less through the shift in the annual mean than through the hot days, heatwaves, and compound heat and drought episodes superimposed on it. Those push organisms past physiological thresholds a slowly warming mean never reaches. Under two degrees of mean warming, a forest acquires more days on which leaf temperature approaches the threshold for photosynthetic damage. It also acquires more nights that fail to relieve the accumulated water deficit, and longer runs of both in succession.

The consequences reach forests through damage as much as through physiology. Extreme drought leaves a measurable imprint on the vitality of individual trees, modulated by their size and their neighbourhood (Rieder et al., 2026). At stand scale, drought-induced dieback and severe wildfire have intensified over recent decades (Cassell et al., 2019), and European disturbance regimes are projected to intensify further under continued warming (Grünig et al., 2026). This matters here because of a circularity. Disturbance removes canopy, and canopy is the physical agent that attenuates the extremes beneath it, so canopy lost to one heat and drought episode cannot buffer the next. Where the canopy is removed outright the buffering does not return quickly: boreal stands need about thirty years after a clear-cut before the sub-canopy temperature is buffered as strongly as in mature forest (Starck et al., 2025).

The relevant question for forest ecology is how hot it gets on the worst days, and where. Near-ground temperature responds to stand structure most strongly in the warm extremes, and average and minimum temperatures differ far less between cut and uncut stands (Potter et al., 2001). The daily maximum is therefore where canopy structure leaves its clearest thermal signature. The where is the half most often left unanswered. Macroclimatic projections are produced on grids of a kilometre or coarser, whereas the conditions that determine whether a seedling survives a heatwave vary over metres.

1.1.2 The forest canopy as a physical modifier of the sub-canopy climate

Forests modify the regional climate they sit in. The canopy intercepts radiation, slows turbulent exchange, and transpires, so the air beneath it is partly decoupled from the free atmosphere above (De Frenne et al., 2019; Geiger et al., 1995). To quantify that decoupling we use a single operational metric throughout this thesis: the canopy temperature offset, $\Delta T_{\text{max}} = T_{\text{micro}} - T_{\text{macro}}$, the difference between the sub-canopy and the above-canopy daily maximum temperature. We focus on the daily maximum because it is the component most relevant to the heat extremes of Section 1.1.1. A negative ΔT_{max} means the understory is cooler than the open air, that is, microclimatic buffering; a positive ΔT_{max} means it is warmer, that is, amplification.

Across biomes, closed canopies buffer: they cool hot daytime maxima and warm cold nights, compressing the temperature range experienced below (De Frenne et al., 2019; Zellweger et al., 2020). The effect is asymmetric between day and night, because its daytime and nighttime causes differ. During the day, foliage intercepts shortwave radiation before it can heat the forest floor, so the understory stays cooler than the open. At night, the same foliage restricts the escape of longwave radiation to a cold sky, so the understory stays warmer. Both effects narrow the diurnal range, but only the daytime one attenuates heat extremes, so ΔT_{max} rather than the mean or minimum offset is the metric of interest here. Old-growth stands, with deep and continuous canopies, buffer most strongly (Frey et al., 2016), and maintaining canopy cover is among the few management options that demonstrably preserve this capacity under future climate (De Lombaerde et al., 2022).

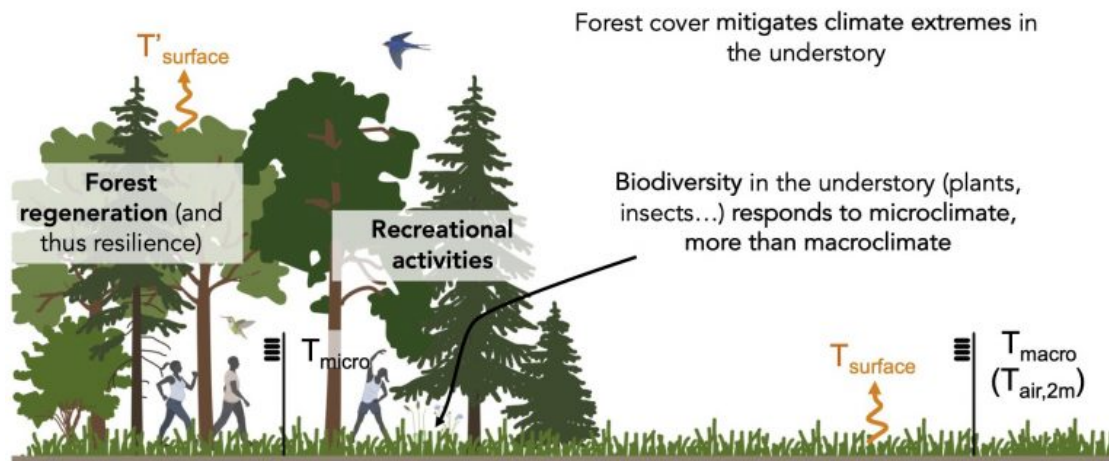


Figure 1.1. The quantity this thesis measures. The canopy separates a sub-canopy temperature from the free-air temperature recorded in the open, and the difference between the two daily maxima is the offset ΔT_{max} used throughout. The understory processes named on the left, forest regeneration and understory biodiversity, respond to the sub-canopy value rather than to the one a weather station records. Figure from the ANR MaCCMic project. **[permission pending: ANR MaCCMic]**

Buffering is not universal. Its sign and strength depend on canopy structure, and where the canopy is sparse or open the offset can reverse, so that the understory overheats relative to the open air (Kovács et al., 2017; Zellweger, Coomes, et al., 2019). Struc-

tural heterogeneity acts in the same direction, widening the diurnal range in stands whose canopy is uneven rather than uniformly closed (Ehbrecht et al., 2019). Severe disturbance can push a stand from the buffering to the amplifying side entirely (Atkins et al., 2023). Species composition and stand density modulate the effect across latitudes, with conifer and broadleaf canopies buffering differently (Díaz-Calafat et al., 2023). In a tree-diversity experiment the effect of species mixing on the offset was largely indirect, acting through the canopy cover a mixture develops (Zhang et al., 2022). Management leaves its own signature on the heterogeneity of buffering within and between stands (Menge et al., 2023). Forest edges add a further gradient, over which sub-canopy conditions converge towards the open within a few tens of metres (Meeussen et al., 2021).

This context-dependent amplification is a documented field phenomenon, observed independently of any model, though radiative overheating of the sensors themselves in open plots can complicate the measurement (Gril, Laslier, et al., 2023). Establishing it on observational grounds matters, because amplification is mistaken for an artifact wherever it later appears in a simulation. Buffering also varies through time, weakening as soils dry (Greiser et al., 2024) and tracking the phenological cycle of leaf area through the season (Zellweger, Coomes, et al., 2019).

Two further qualifications bound what this thesis claims. First, temperature is not the only variable buffered. The canopy modifies atmospheric humidity, and with it vapour pressure deficit, at least as strongly as it modifies temperature, and vapour pressure deficit is more proximate to plant water stress (Burton et al., 2024). We lead on temperature because it is what the logger networks record, and therefore the only quantity against which simulations can be validated here. Where a vapour-pressure-deficit offset is computed, it remains model-internal for want of a matching observation. Second, canopy structure is not the only driver. Topography redistributes radiation and cold air, soil moisture modulates the evaporative term, and the two interact with structure rather than adding to it (Davis et al., 2019; Jucker et al., 2018). The microclimate work of this thesis is conducted at a lowland site of low relief, so structure can be read with less topographic confounding than at a site of strong relief.

1.1.3 Why fine-scale temperature variability matters

Fine-scale variability is the climate understory organisms experience. Sub-canopy temperature governs the composition and dynamics of understory plant communities, and buffered forests can slow the thermophilisation of their flora under macroclimate warming (De Frenne et al., 2013; De Pauw et al., 2022). When species distributions are re-examined against microclimatic rather than macroclimatic temperatures, the thermal niches inferred for forest plants shift by several degrees (Haesen, Lenoir, et al., 2023). Much of the published climate-change vulnerability of forest understory species has therefore been estimated against the wrong temperature. Vascular plants and bryophytes differ in their affinity for buffered conditions, so the microclimate filters communities rather than shifting them wholesale (Gril et al., 2024).

Regeneration is the second process at stake, and it is the one most directly tied to canopy structure. Seedling survival and early growth respond to the temperature and moisture of the first metre above the ground, not to the conditions logged at a weather

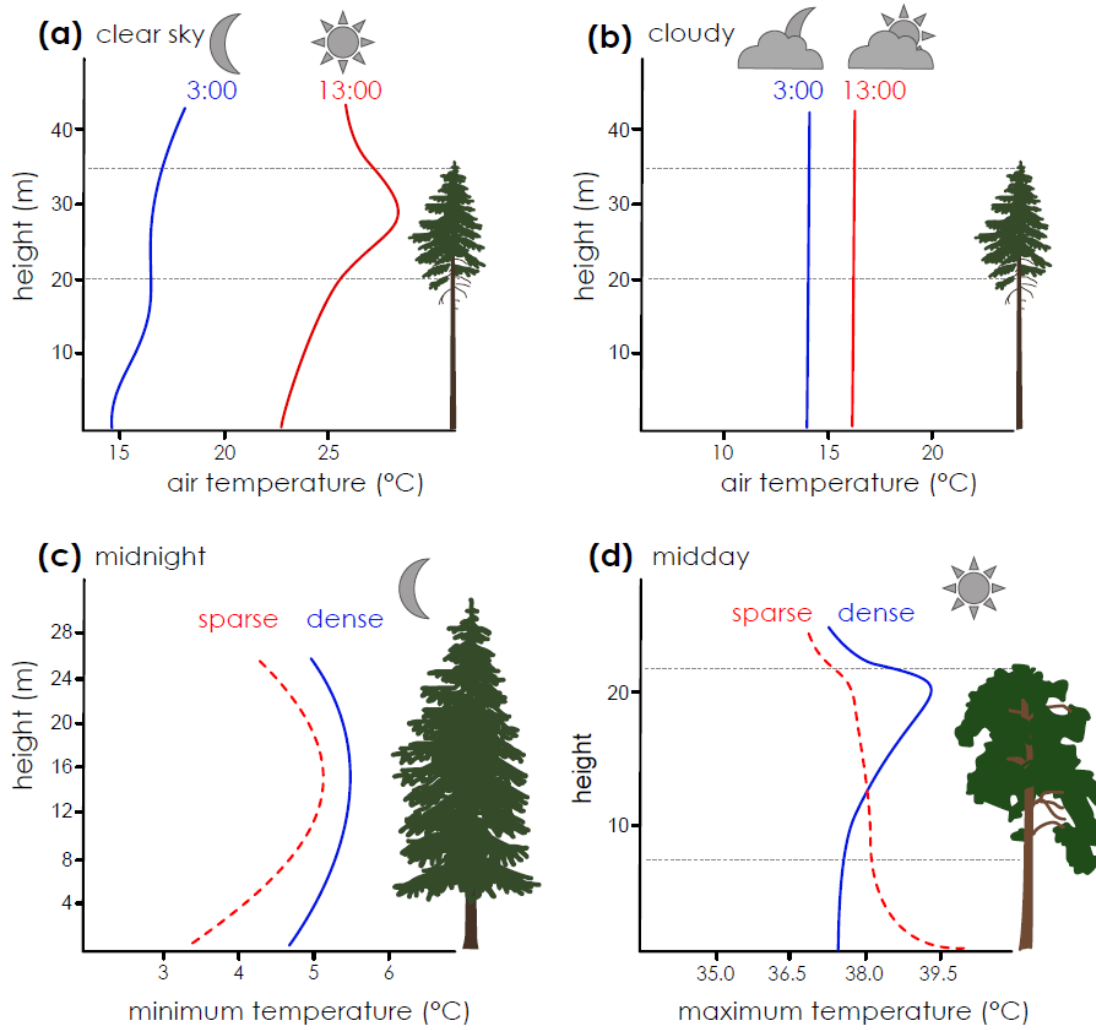


Figure 1.2. Vertical air temperature profiles inside forests of contrasting canopy structure. Panels (a) and (b) contrast a clear-sky and an overcast day: the daytime profile develops a maximum inside the crown only when the sun is out, and under cloud the column is nearly isothermal. Panels (c) and (d) contrast a sparse and a dense canopy of the same height at night and at midday, and they carry the asymmetry this thesis measures. At midnight the sparse stand is the colder of the two through the whole column; at midday it is the warmer one near the ground, while the dense stand holds its maximum up in the crown and keeps the understory below it cooler. Buffering and amplification are therefore the same physics read at two times of day, and the sign depends on how much foliage sits above. Reproduced from De Frenne et al. (2021), Figure 4. [permission pending: Wiley]

station kilometres away (Meeussen et al., 2022; Von Arx et al., 2013). Juveniles and adults of the same species do not experience the same climate, and the thermal gap between them has widened over recent decades (Caron et al., 2021). Since the juvenile stage occupies the sub-canopy layer that the overstory buffers, the demographic bottleneck of temperate forests is set in the sub-canopy layer. Fauna, too, exploit the structure: canopy-derived thermal shelters are measurably used by large mammals during summer heat (Melin et al., 2014).

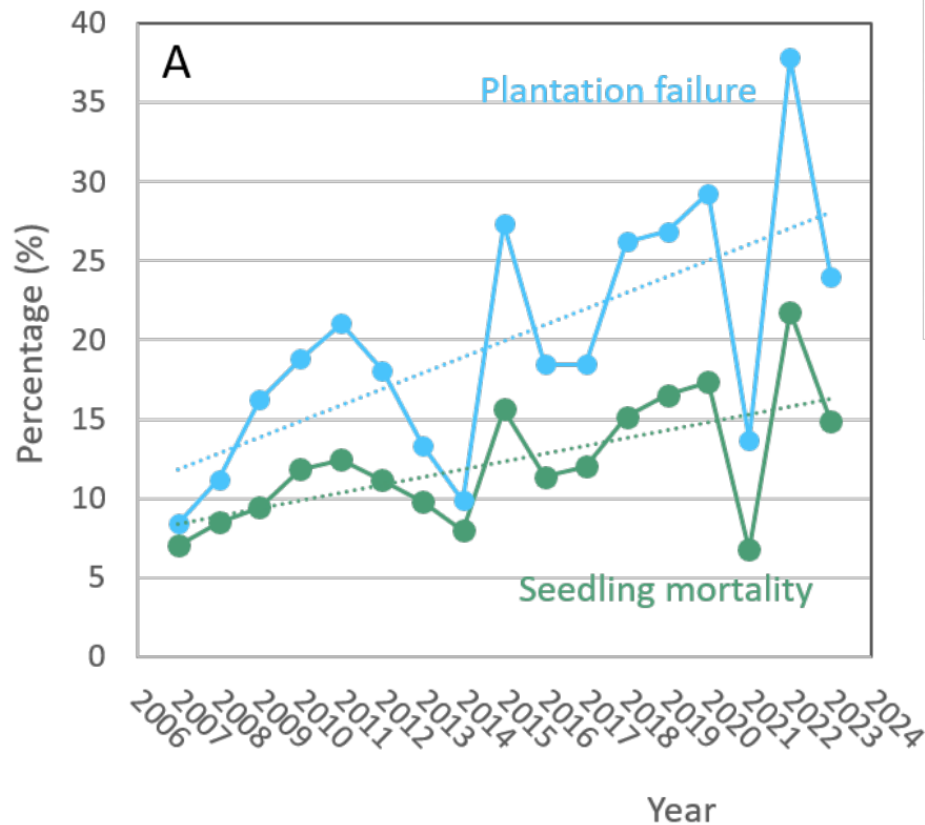


Figure 1.3. Plantation failure and seedling mortality in French forests, 2006 to 2023. Both rise over the period, and both spike in the years that followed the 2018 to 2022 droughts: plantation failure reaches nearly 38 % in 2022 against about 8 % at the start of the record. The figure shows why the temperature of the first metre above the ground is not an academic quantity. Regeneration is the demographic bottleneck of temperate forest, it happens in exactly the layer the overstory buffers, and it is already failing at rates that a stand-scale average temperature does not explain. Panel A of a figure adapted by Bouwen (2025) from the French forest health service. **[permission pending: DSF / Bouwen 2025]**

Under a warming climate, structurally buffered stands can act as thermal microrefugia, pockets where cool-adapted species persist while the surrounding region warms (Lenoir et al., 2017; Lenoir et al., 2022). Accounting for them materially changes the velocity at which species must move to track a shifting climate (Soifer et al., 2026), and microclimatic heterogeneity within a stand is itself associated with structural complexity and with biodiversity (Ehbrecht et al., 2017). Beyond the species level, the microclimate modulates ecosystem functioning, mediating part of the effect of plant diversity

on process rates (Beugnon et al., 2024). Ignoring the sub-canopy layer therefore biases our expectations of biodiversity change (Kemppinen et al., 2024; Lembrechts et al., 2019). Whether such refugia can be located and mapped depends on our ability to predict the microclimate, not only to describe it in a few places (Zellweger et al., 2020).

That dependence sets the standard the rest of this work has to meet. A microclimate prediction is only as good as the canopy description it is given, and the errors that matter are structured. A structural input that under-reports foliage in dense canopy will make buffered stands look less buffered than they are. One that fails to register the openness of a gap will hide the amplification that determines seedling survival. Both mistakes compress the range of conditions a forest contains, so a model given such an input can fit the observations on average while misplacing the extremes that decide where a microrefugium lies.

1.1.4 Observing microclimate: logger networks and their spatial limitation

The empirical foundation of microclimate ecology is the in-situ temperature logger. Networks of small sensors, typified by HOBO loggers and coordinated efforts such as SoilTemp, now supply large, standardised archives of sub-canopy conditions (Lembrechts et al., 2022; Lembrechts & Lenoir, 2020). They are accurate, they resolve the daily cycle, and every simulation reported in this thesis is answerable to them (Maclean et al., 2021).

Deploying them well takes care. A sensor exposed to direct sunlight reads its own radiative load rather than the air temperature. Shielding, orientation, and mounting height are therefore part of the measurement, and the bias is worst in the open plots where amplification is expected (Gril, Laslier, et al., 2023; Maclean et al., 2021). The placement of loggers determines which part of the structural gradient is sampled, and a network that oversamples closed canopy will underestimate the range of conditions present (Lembrechts et al., 2021). Dense local networks can be interpolated into high-resolution grids where the instrumentation effort is sufficient (Brůna et al., 2026), but the effort scales with area.

The core limitation is simpler than any of these: a logger measures one point. It cannot, on its own, map how $\Delta T_{\max}$ varies continuously across a heterogeneous forest landscape, because the sensor network is always far sparser than the structural variation it is trying to capture (Zellweger, De Frenne, et al., 2019). A campaign of fifty loggers in a forest of several thousand hectares samples a vanishing fraction of the structural configurations present, and the configurations it misses are not random.

Predicting the microclimate between the points is therefore the central problem of this thesis. It means combining two things: a plot-scale mechanistic understanding of how canopy structure controls temperature, and continuous remotely sensed descriptions of that structure. Every sensor imposes a trade-off. Airborne LiDAR resolves three-dimensional structure but only once and locally, whereas satellite optical imagery is repeated but two-dimensional and prone to saturation. The three research chapters address it in turn.

1.2 Drivers of forest microclimate and the role of canopy structure

1.2.1 Linking three-dimensional vegetation structure to sub-canopy microclimate

Buffering is an energy-balance phenomenon before it is an ecological one. What reaches the forest floor, and what the sub-canopy air does with it, is set by four coupled processes, each of which depends on where the foliage sits.

The first is shortwave interception. Solar radiation is attenuated as it passes through the canopy, approximately exponentially with cumulative leaf area, so the energy available to heat the understory falls as the canopy thickens. The rate of that attenuation is governed by an extinction coefficient, which is not a universal constant. It depends on leaf inclination, on solar zenith angle, and on how foliage clumps rather than distributing itself randomly (D. Baldocchi et al., 1984; Campbell, 1986). Two canopies with identical leaf area but different leaf angle distributions therefore transmit different amounts of light. The second process is the longwave exchange that partly reverses the first at night. Foliage emits downward and screens the understory from a cold sky, so closed stands are warmer than the open after dark. The buffering of minima and of maxima are therefore physically distinct phenomena rather than two views of one.

The third is turbulent transfer. Foliage exerts aerodynamic drag, so wind speed decays with depth into the canopy and exchange with the atmosphere above weakens, leaving the sub-canopy air partly insulated from whatever the free atmosphere is doing (Geiger et al., 1995). The consequence is not uniformly cooling. Weak mixing insulates the understory from warm air aloft, but it equally traps whatever heat is generated below. A sparse canopy that intercepts little radiation while still suppressing ventilation can therefore amplify rather than buffer. Representing this exchange correctly within and just above the canopy has proved one of the harder problems in land surface modelling (Bonan et al., 2018).

The fourth term is latent heat. Every unit of energy consumed by evaporation is a unit that does not warm the air. Canopy transpiration is only part of it: the understory vegetation and the soil surface evaporate as well, and where soil water remains available that flux removes part of the daytime energy from the sensible-heat budget. Transpiration in temperate oak stands tracks leaf area closely and varies strongly between years with water availability (Bréda & Granier, 1996), and understory leaf area is itself a non-negligible and poorly measured quantity (George et al., 2021). Buffering consequently weakens as the local water balance dries, so the same canopy buffers less in a drought year than in a wet one (Davis et al., 2019; Greiser et al., 2024).

Two smaller terms close the balance. Heat conducted into and out of the soil buffers the near-ground air on its own account, storing energy during the day and releasing it at night. Soil and air offsets are therefore correlated but not interchangeable, and logger networks increasingly record both (Lembrechts et al., 2022). The energy partitioning is expressed in humidity as much as in temperature. The same latent flux that fails to warm the air does raise its vapour content, so temperature buffering and humidity buffering are two readings of one partitioning (Burton et al., 2024).

Canopy models usually collapse the understory layer and the soil surface into a lower boundary condition rather than resolving them with the detail given to the overstory.

Energy a model fails to route into evaporation it routes into warming the air instead.

Each of these four processes is height-dependent, so structure is three-dimensional rather than scalar. Radiation is intercepted, drag is exerted and transpiration occurs where the foliage is, so the canopy heat source sits at whatever height the foliage is concentrated. Detailed three-dimensional descriptions of old-growth canopies were built to reason about this coupling between architecture, radiation balance, and gas exchange (Parker et al., 2004). Modelling studies indicate that vertical canopy architecture affects transpiration and leaf thermoregulation at fixed total leaf area (Banerjee & Linn, 2018). Take two stands holding the same total leaf area, one with foliage packed near the top and one with it spread evenly down the profile. They do not present the same vertical arrangement of sources and sinks to the air beneath. Whether that difference translates into a measurable difference in ΔT_{max} , and how large it is relative to having more or fewer leaves, is the question the next section poses.

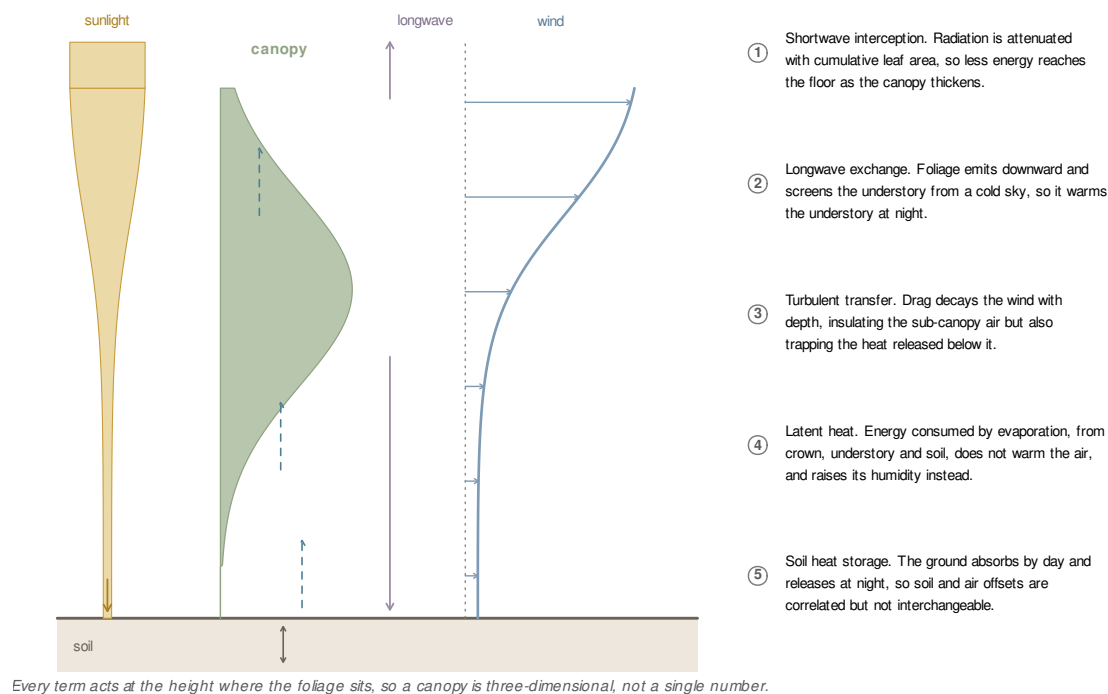


Figure 1.4. Schematic: the four processes that set the sub-canopy temperature, and the heights at which they act. Shortwave radiation is intercepted as it passes through the foliage, longwave exchange partly reverses that at night, aerodynamic drag decays the wind with depth, and evaporation consumes energy that would otherwise warm the air. Soil heat storage closes the balance. Each term acts where the foliage sits, so a canopy has to be described in three dimensions rather than by a single number. No axis carries measured values.

1.2.2 Which dimension of structure? Leaf quantity and vertical arrangement

Canopy structure is usually summarised by the leaf area index (LAI), the one-sided leaf area per unit ground area (Chen & Black, 1992). LAI is attractive because it is a

single number, because it enters radiative transfer directly, and because it correlates strongly with observed buffering across gradients from closed forest to open plantation (Hardwick et al., 2015; Von Arx et al., 2013). Even the term requires care. Most instruments and most remote-sensing retrievals return an effective LAI that assumes a random distribution of foliage, and so underestimate true leaf area wherever the canopy is clumped (Chen, 1996; Chen & Black, 1992). Optical instruments and gap-fraction inversions return plant area, which includes woody elements, rather than leaf area alone (Brown et al., 2024). Radiative-transfer schemes differ in whether they expect a one-sided or a two-sided leaf area. Comparisons between sensors are only meaningful once these conventions are aligned.

Table 1.1 fixes the conventions this thesis uses, and names what each retrieval returns.

Table 1.1. The vocabulary of leaf area, and what each retrieval returns. The distinctions are not pedantic: a large part of the disagreement between published leaf area products is definitional before it is physical. Two consequences run through the thesis. The extinction coefficient of the airborne inversion is a stated choice rather than a measured property. It rescales the retrieval by a constant, so the absolute magnitude depends on it while the ranking between plots does not. No row of this table is a field measurement of leaf area, so every comparison reported here is between two model-based estimates.

Quantity	What it denotes	What is available here
True leaf area index	One-sided leaf area per unit ground area (Chen & Black, 1992)	Not measured. No coincident field measurement of leaf area exists at any of the three sites, so the comparisons of Chapter 2 are posed as inter-sensor consistency rather than accuracy.
Effective plant area index, from ALS	The vertically integrated one-sided plant area density, obtained by inverting the airborne gap fraction through the Beer-Lambert law	The structural leaf area of Chapters 1 and 3, and one side of the comparison in Chapter 2. It counts woody surface, it assumes a spherical leaf-angle distribution through a fixed extinction coefficient, and it applies no clumping correction, so it is a proxy for true leaf area rather than a measurement of it.
Plant area density profile	The vertical density of that plant area, layer by layer	Retrieved from the same point cloud, and its integral is the index above. This is the quantity a multilayer canopy scheme can accept and a single index cannot express, which is the distinction Section 1.2.2 turns on.

Quantity	What it denotes	What is available here
Fractional cover	The share of ground the canopy covers	Computed as the fraction of airborne returns above a 2 m height threshold within the footprint. It enters the canopy model as the clumping factor: it changes how efficiently a given leaf area intercepts radiation, not how much leaf area the column holds.
Optical leaf area index, from Sentinel-2	The leaf area returned by inverting a canopy radiative-transfer model against the observed reflectance	The satellite leaf area of Chapters 2 and 3. It is conditioned by the prescribed parameter ranges that regularise the inversion, and its sensitivity is exhausted in dense canopy (Section 1.4.2).
Two-sided leaf area	Total leaf surface per unit ground area, twice the one-sided value	What the multilayer model reads. The one-sided airborne index is doubled before input. This is a units convention, not a clumping correction, and conflating the two is the most common way the comparison goes wrong.

Leaf area is also not a constant of a stand. It follows a phenological cycle, rising through leaf-out, plateauing through summer, and declining through senescence. In deciduous forest the amplitude of that cycle exceeds most of the between-stand variation observed at any single date (Bréda & Granier, 1996; Cotrina-Sanchez et al., 2026). A single-date structural description is therefore a snapshot of a moving quantity, which is one reason the temporal signature of each sensor, developed in Section 1.4, is not a secondary consideration.

LAI is also an integral, and integrals discard information. The same LAI can be realised by a dense canopy covering part of the ground or by a thinner canopy covering all of it. It can equally be realised by foliage concentrated in a narrow upper crown layer or distributed through a deep, multi-layered profile. Several complementary descriptors have therefore been proposed. Fractional cover (fCover) captures the horizontal dimension, maximum height ($H_{\max}$) the depth of the column available for attenuation, and the vertical profile of leaf area density (LAD) the arrangement within that column. The idea of characterising a canopy by its foliage profile is old (MacArthur & Horn, 1969), but it became routinely measurable only with laser scanning (Arnqvist et al., 2020; Kamoske et al., 2019). A parallel literature has developed indices of structural complexity that compress the three-dimensional point cloud into scalars (Beckschäfer et al., 2013; Kane et al., 2010; McElhinny et al., 2005). A further strand distinguishes the horizontal from the vertical component of heterogeneity, and the two can act in opposite directions on ecological responses (Carrasco et al., 2019).

The literature has not settled which of these dimensions governs the thermal effect. Some studies report canopy density or cover as the dominant control (Kovács et al., 2017; Zellweger, Coomes, et al., 2019). Others find that structural complexity and

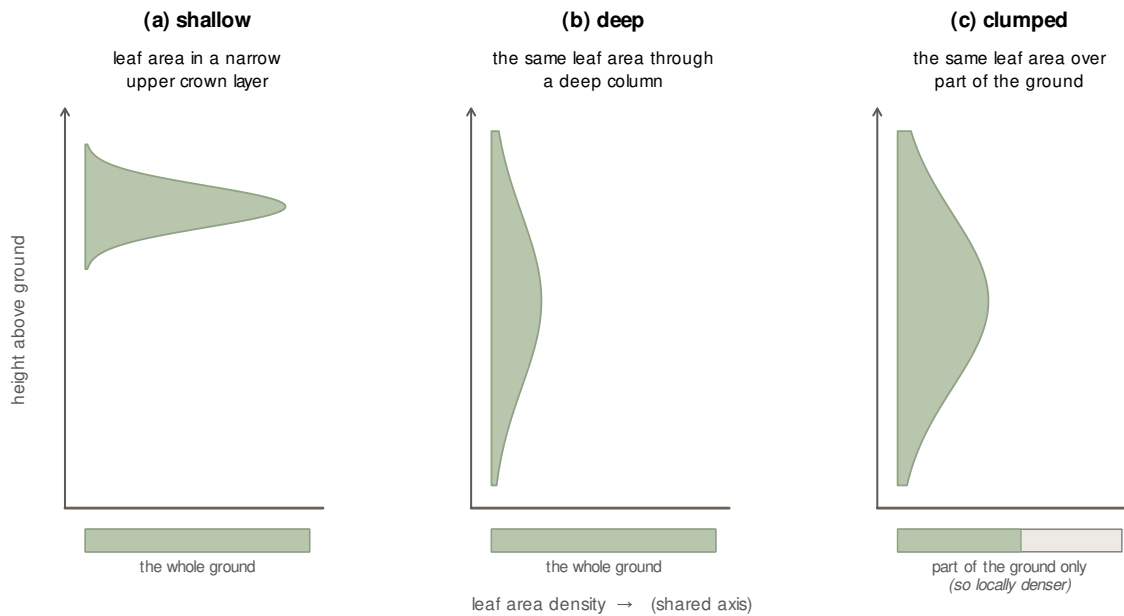


Figure 1.5. Schematic: leaf area index is an integral, and integrals discard information. Three canopies with the same leaf area index per unit ground. In (a) and (b) the shaded areas are equal and only the vertical arrangement of the foliage differs; in (c) the same leaf area is gathered over part of the ground, so the profile is locally denser. A single number cannot separate the three, although they present very different surfaces to radiation and to the air beneath. No axis carries measured values.

vertical heterogeneity carry independent explanatory power for the diurnal temperature range (Ehbrecht et al., 2017; Ehbrecht et al., 2019). Others again show the two changing together along disturbance and land-use gradients (Atkins et al., 2023; Jucker et al., 2018). In boreal stands scanned with terrestrial LiDAR, the total quantity of plant material was not by itself an adequate predictor of the buffering. The number of canopy layers and the density of the understory carried it instead (Starck et al., 2025). The same ambiguity recurs outside the microclimate literature, where canopy structural complexity has been proposed as a driver of productivity in ways that are equally hard to separate from leaf quantity (Fahey et al., 2019; Hardiman et al., 2011).

The disagreement has a structural cause. In real stands these variables covary: tall stands tend to be dense, dense stands tend to be closed, and closed stands tend to have deep multi-layered profiles. A study that finds LAI dominant and a study that finds vertical complexity dominant may be reading the same underlying gradient through different predictors. No amount of additional field data collected along that gradient will separate them. We therefore treat the question as open, in the terms this thesis can test: does the total quantity of foliage govern sub-canopy buffering, or does its vertical arrangement?

1.3 Modelling the forest microclimate

1.3.1 Statistical versus mechanistic approaches

Two families of models predict sub-canopy temperature, and they fail in different places. Statistical, or correlative, models regress the observed offset on structural and topographic predictors, using linear models, generalised additive models, or machine learning trained on logger networks. Monthly microclimate models have been built for managed boreal landscapes (Greiser et al., 2018), and fine-scale warming rates have been mapped from terrain and canopy predictors (Maclean et al., 2017). Generalised additive models interpolate microclimate within stands (Burnett & Anderson, 2019). Continental gridded products such as ForestTemp and ForestClim now provide sub-canopy bioclimatic variables for European forests (Haesen et al., 2021; Haesen, Lembrechts, et al., 2023), alongside global maps of soil temperature (Lembrechts et al., 2022). Parsimonious formulations that summarise the micro-macro relationship by a slope (the buffering slope, as we call it throughout) and an equilibrium term have proved well suited to comparing sites (Gril, Spicher, et al., 2023). Within the range of conditions they were fitted on, all of these interpolate well.

Their limitation is the question of Section 1.2.2. When predictors are strongly collinear, a fitted coefficient is not an effect. Regression coefficients become unstable and their standard errors inflate. The additive analogue in generalised additive models is concavity, in which one smooth term can be partly reproduced by the others, so the decomposition into individual contributions is not identifiable (Dominici, 2002). Machine learning relocates the problem: permutation-based variable importance is unstable under predictor correlation, and correlated predictors share importance in ways that depend on the algorithm rather than on the data (Gregorutti et al., 2017; Nicodemus et al., 2010). A model of this kind can rank plots correctly while attributing the ranking to the wrong variable. It is also site-fitted and descriptive, so it extrapolates poorly to structural combinations absent from its training data. It offers no mechanism that

would separate the radiative role of leaf quantity from the turbulent role of leaf arrangement.

Mechanistic models take the opposite route. Rather than fitting the offset, they solve the coupled energy, water, and radiative balance that produces it, so their parameters are in principle transferable rather than calibrated to one site (Maclean & Klings, 2021). The family now includes dedicated microclimate models (Maclean & Klings, 2021), validated against observations in temperate understories (Brusse et al., 2024) and applied to fine-scale variability in boreal forests (Kolstela et al., 2024). It also includes radiative-transfer approaches to microclimate mapping (Zellweger et al., 2024) and land surface models applied to projections of future forest microclimate (Hes et al., 2024). Mechanistic and empirical approaches have also been compared directly in degraded tropical forest, with neither dominating on accuracy alone (Marsh et al., 2022).

A related difficulty concerns what “accurate” should mean for a microclimate model, and it governs how any of them can be compared. A model can be judged on how close its simulated offset is to the observed one. It can also be judged on whether it orders plots correctly, or on whether it reproduces the observed spread between them. A model carrying a constant warm bias may rank plots perfectly while being wrong everywhere in absolute terms. A model that compresses the simulated range may look accurate on average while being unable to distinguish a buffered plot from an amplifying one. The ecological use of a microclimate map is usually to find the cool places rather than to predict an exact temperature. This thesis therefore reports ranking, spread and bias separately throughout, never collapsed into one score.

This thesis needs a mechanistic model for controllability, not for accuracy. Because the canopy description enters as an input rather than as a fitted covariate, we can change leaf quantity while holding arrangement fixed, and change arrangement while holding quantity fixed. The collinearity that makes attribution impossible in observational data is absent from a designed set of simulations, because the experimenter, not the forest, decides which combinations are run.

1.3.2 The MuSICA model

MuSICA is a multilayer soil-vegetation-atmosphere transfer model that solves energy, water, and carbon exchange through a vertically discretised canopy (Ogée, Brunet, et al., 2003). Radiative transfer is computed layer by layer, and separately for sunlit and shaded foliage. The leaf energy balance and stomatal conductance are solved at each level, and turbulent transfer links the layers to the air above. Soil water and heat are treated in a coupled multilayer column. The model was developed and evaluated at European forest flux sites across timescales from hourly to yearly (Ogée, Brunet, et al., 2003; Ogée, Peylin, et al., 2003). It has already been coupled to LiDAR-derived structure to study sub-canopy microclimate in conifer stands (Bouwen, 2025).

That study is the closest existing work to this one. It established that a multilayer canopy model can reproduce the ordering of sub-canopy conditions across stands of contrasting density. It also established that the meteorological forcing must itself be corrected for canopy structure before scenarios of differing structure can be compared.

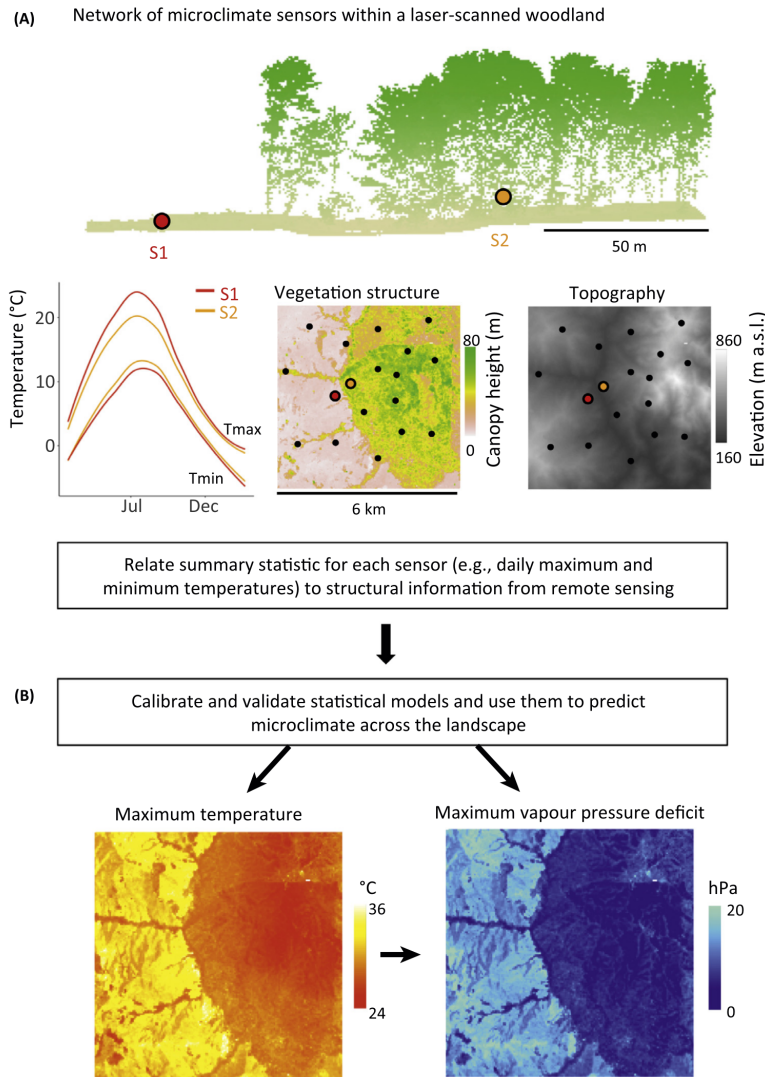


Figure 1.6. The statistical route to a microclimate map, which is the approach this thesis departs from. **(A)** A network of loggers is placed inside a laser-scanned woodland, each sensor is summarised into an ecologically meaningful statistic such as a daily maximum or minimum, and those statistics are related to remotely sensed descriptors of vegetation structure and topography. **(B)** The fitted statistical model is then used to predict microclimate across the landscape. The route works, and it produces the maps that motivate the field. What it cannot do is say which structural variable causes the pattern, because the descriptors entering the fit covary in the stands that were sampled, which is the gap Section 1.5.1 sets out. Reproduced from Zellweger, De Frenne, et al. (2019), Figure 1. **[permission pending: Elsevier]**

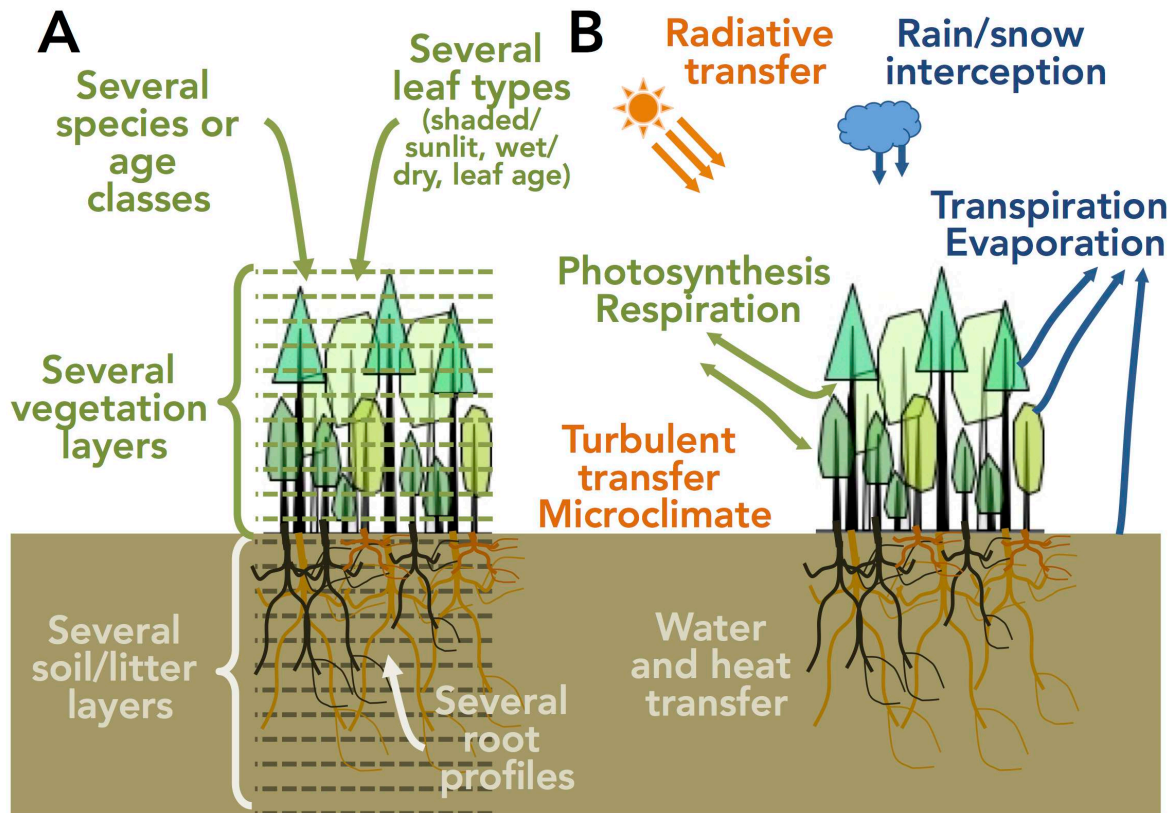


Figure 1.7. What the MuSICA model contains. **(A)** The discretisation: several vegetation layers, several species or age classes, several leaf types within a layer, sunlit and shaded, wet and dry, and several soil and litter layers with their own root profiles. **(B)** The processes solved on that discretisation: radiative transfer, rain and snow interception, photosynthesis and respiration, transpiration and evaporation, turbulent transfer, and the coupled transport of water and heat in the soil. Sub-canopy microclimate is an output of the turbulent and radiative terms rather than a separate module, which is what makes the model answerable to a temperature logger. Figure by Jérôme Ogée, author of MuSICA (Ogée, Brunet, et al., 2003). **[permission pending: J. Ogée]**

For lack of measured vertical profiles, the distribution of foliage was described throughout by a parametric beta function. That unimodal shape cannot represent the stratified canopies produced by partial harvesting, and its use was judged to have contributed to an underestimation of the simulated buffering. Its closing recommendation was to feed measured LiDAR-derived profiles of plant area density into the model in place of that function.

Three things remain open. The measured profile has not been substituted for the assumed one, so the effect of the vertical arrangement itself has not been isolated. A profile held to a single parametric family cannot be contrasted against an alternative arrangement at matched leaf area and height. The work was conducted in evergreen conifer stands, where the seasonal cycle of foliage and the geometry of interception both differ from a temperate deciduous canopy. And nothing in it addresses how the structural input might be obtained beyond the footprint where the LiDAR was flown, the separate problem taken up in Section 1.3.3.

Three properties make MuSICA the appropriate tool for the question of Section 1.2.2. The first is that it is multilayer. A vertical profile of leaf area density is a native input, whereas a big-leaf scheme collapses the canopy into a single effective surface. If the arrangement of foliage is to be tested against its quantity, the model must be able to see both. MuSICA belongs to a small family of canopy schemes with this property: multilayer biophysical models developed for temperate deciduous stands (D. D. Baldocchi et al., 2002), three-dimensional radiation and gas-exchange models (Sinoquet et al., 2001), and integrated soil-canopy radiative and energy-balance models (Van Der Tol et al., 2009). Multilayer canopy parameterisations are now being introduced into land surface models as well (Bonan et al., 2014; Bonan et al., 2018). Empirical microclimate downscalers take canopy cover or a height metric as a scalar predictor and could not accept a profile even if one were available.

The second is that the coupling between structure and temperature is physical rather than empirical. No transfer function has to be fitted between the two, which removes the attribution problem of Section 1.3.1. A sensitivity computed by perturbing an input of a physical model is a property of the physics, not an estimate contaminated by the covariance structure of a training set.

The third is that the canopy inputs are explicit and separable. Leaf area, canopy height, cover, and the shape of the vertical profile enter as distinct quantities. That is what makes controlled perturbation possible, and what allows the vertical profile to be contrasted against a leaf-area- and height-preserving uniform alternative.

One limitation bounds everything that follows. MuSICA is a one-dimensional column, so lateral advection, edge effects, and sub-pixel canopy gaps lie outside its representation. No such model can be expected to reproduce a plot whose surroundings differ from that column, for instance one containing a gap the mean structural input does not resolve. The argument developed here therefore rests on relative comparisons between plots rather than on absolute temperatures.

1.3.3 From plot to landscape

Mechanistic canopy models are plot-scale by construction. They are one-dimensional columns: they require, for every location simulated, a complete structural description

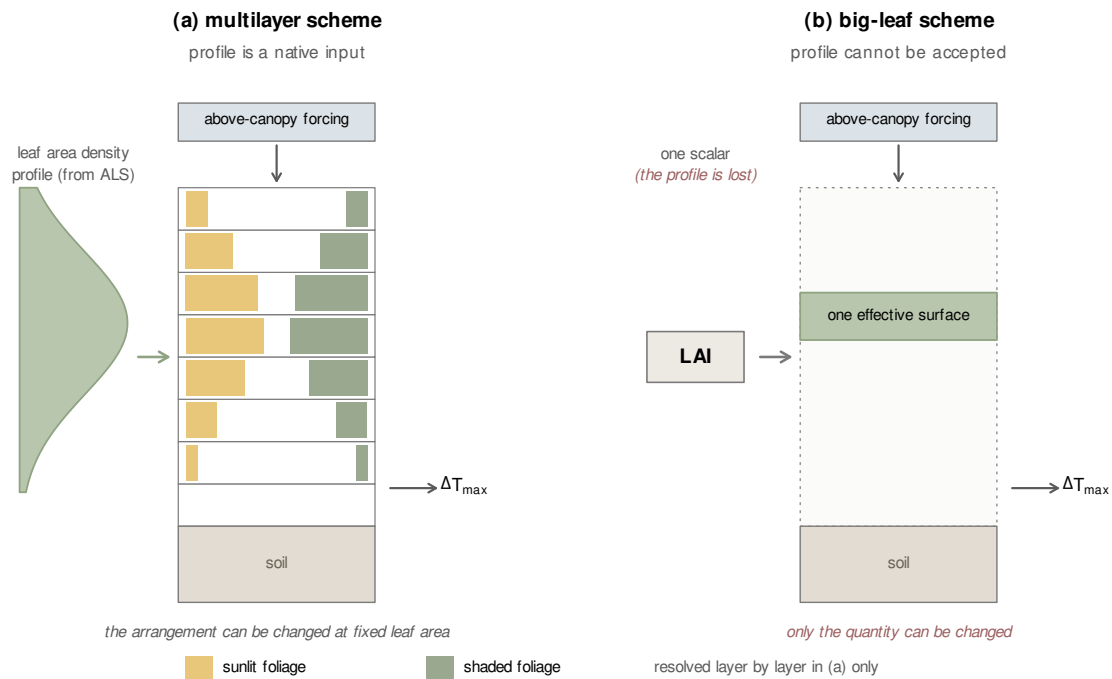


Figure 1.8. Schematic: why the canopy scheme has to be multilayer. What matters here is not what the model computes but what it can accept. In (a) a vertical profile of leaf area density is a native input: the canopy is discretised into layers, each holding sunlit and shaded foliage with its own energy balance, so the arrangement of that leaf area can be altered while its total is held fixed. In (b) the column is collapsed into one effective surface, which can only be given a scalar; the profile is lost on the way in, and the question of Section 1.2.2 cannot be posed at all. Layer count and profile shape are illustrative and carry no measured values.

of the canopy above that point together with the meteorological forcing above it. Running one such column is cheap.

The structural inputs are the bottleneck. Measuring LAI in the field, whether by hemispherical photography, optical plant-canopy analysers, or destructive sampling, is slow, weather-dependent, and subject to its own methodological uncertainty (Chen et al., 1991; Holst et al., 2004; Weiss et al., 2004). Different indirect methods disagree with one another even when applied at the same sites, and the disagreement is largest where it matters most, in the understory layer (George et al., 2021). Automated processing has improved reproducibility (Brown, Morris, Leblanc, et al., 2023), but it has not changed the arithmetic. Measuring a vertical LAD profile in the field is harder still, and a campaign of that kind yields tens of plots, occasionally hundreds.

A landscape contains millions of model pixels. There is no route from plot-scale mechanism to landscape-scale microclimate map that does not pass through a structural dataset acquired remotely, continuously, and at fine spatial resolution.

Choosing that resolution is a modelling decision, not a technical one. A one-dimensional column assumes that the canopy above the simulated point is horizontally uniform over the area it represents. The pixel must therefore be small enough for that assumption to hold and large enough for the structural retrieval to be reliable. Too coarse, and a single column has to stand for a mixture of gaps and closed canopy whose average behaves like neither (Garrigues et al., 2006). Too fine, and the retrieval noise of the structural input dominates the structural signal. Attributing a thermal effect to canopy structure calls for footprints large enough to define a stable vertical profile, while comparing satellite against airborne retrievals is constrained to the 10 m grid the satellite imposes. Both sit within the range over which sub-canopy temperature has been shown to vary in temperate stands (Menge et al., 2023; Zellweger, De Frenne, et al., 2019). Which instrument can populate that grid, and what each one costs in structural fidelity or temporal coverage, is the subject of the next section.

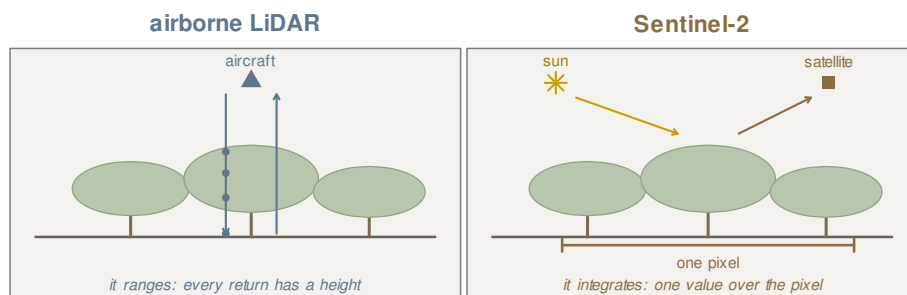
1.4 Remote sensing of forest structure: bridging the spatial scale

1.4.1 Airborne LiDAR

Airborne laser scanning (ALS) is the only operational technique that resolves the interior of a forest canopy over an area. Laser pulses penetrate gaps in the foliage and return a three-dimensional point cloud. The ground surface is identified and the cloud normalised to height above ground. The vertical gap fraction is then computed layer by layer and inverted, through a Beer-Lambert formulation, into a profile of plant area density and an integrated LAI (Bouvier et al., 2015; Richardson et al., 2009; Zheng et al., 2013). The approach has been established for two decades and validated across forest types and sensors (X. Chen et al., 2004; Kwak et al., 2010; Lefsky et al., 2002; Morsdorf et al., 2006; Peduzzi et al., 2012). Reconstructing full LAD profiles rather than integrated totals is now routine (Arnqvist et al., 2020; Kamoske et al., 2019; Lin & West, 2016).

The retrieval is not parameter-free, and its assumptions matter for this thesis. The conversion from gap fraction to leaf area rests on an extinction coefficient that encodes leaf angle distribution and is commonly fixed at a nominal value. It also assumes a

how one
acquisition
works



energy source	its own laser pulse	sunlight the canopy reflects
what is measured	the time each return takes to come back	radiance in each spectral band
one acquisition returns	many ranged returns per pulse, in three dimensions	one value per band over a 10 or 20 m pixel
what limits it	the flight plan, the pulse density, the cost	cloud cover, and the sun and view angles

one instrument is active and resolves height; the other is passive and resolves colour

Figure 1.9. Schematic: the two sensors as instruments. Airborne LiDAR carries its own light and times the return of each interception, so a single acquisition yields many returns whose heights are known. Sentinel-2 collects sunlight the canopy reflects and records one radiance value per spectral band over a 10 or 20 m footprint, so a single acquisition yields no height at all. Everything the following sections compare, the vertical profile on one side and the saturating two-dimensional signal on the other, follows from that difference. No axis carries measured values.

random foliage distribution, which real canopies violate through clumping (Hu et al., 2018). The returns come from wood as well as leaves, so the retrieval is a plant area index unless the two are separated, which matched leaf-on and leaf-off acquisitions can do in deciduous stands (Zhu et al., 2020). Acquisition geometry leaves its own imprint, since flying height, scan angle, and pulse density all affect the signal’s penetration depth. Uncertainty in voxel-scale plant area density therefore has to be propagated rather than ignored (Pimont et al., 2018), and canopy height models derived from the same cloud carry their own processing choices (Khosravipour et al., 2014). A LiDAR LAI is a model-based estimate resting on stated assumptions, not a direct measurement. That matters as soon as such an estimate is set against an optical one, because neither side of the comparison is a reference truth.

ALS also delivers more than leaf area. The same point cloud yields a digital terrain model, a canopy height model, fractional cover, and the full suite of height and density metrics on which forest inventory modelling rests (Bouvier et al., 2015; Khosravipour et al., 2014). An ALS survey is best read as a structural description of a stand rather than as a leaf-area sensor. Canopy height is easier to retrieve than leaf area and is correlated with it, which makes it the fallback wherever a full profile is unavailable. Terrestrial and portable scanning reconstruct leaf area density at the scale of individual crowns (Hosoi & Omasa, 2006; Li et al., 2017), but they cover plots rather than landscapes.

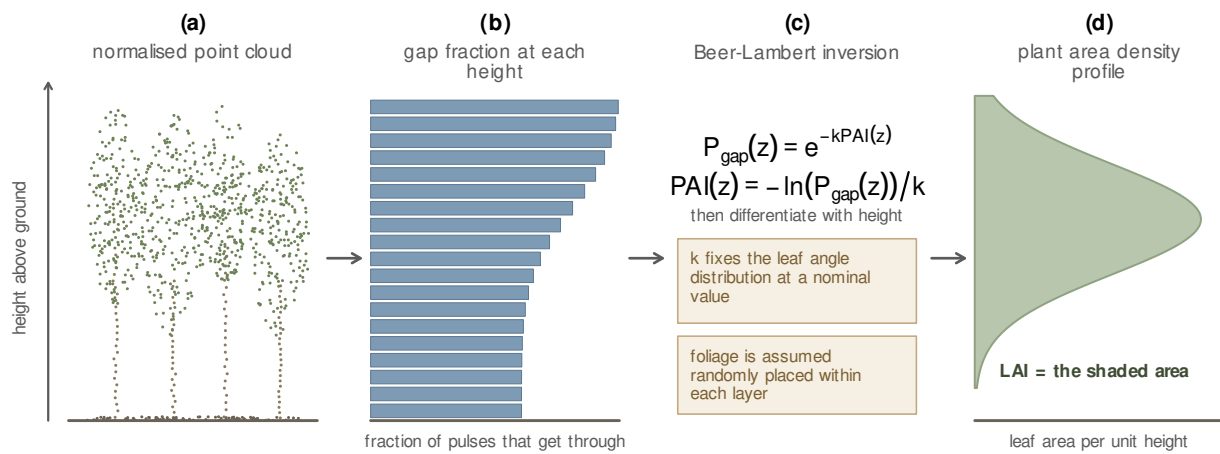
ALS-derived structure has since become a standard input to microclimate studies, whether to map buffering directly (Gril, Laslier, et al., 2023; Vandewiele et al., 2023), to constrain it jointly with topography (Jucker et al., 2018), or to explain the thermal shelters used by wildlife (Melin et al., 2014). For the question of Section 1.2.2 its value is specific: it is the sensor that delivers the vertical arrangement, not a proxy for it.

Its spatio-temporal signature is the mirror image of that strength. ALS is acquired by aircraft campaign, which makes it expensive, spatially bounded by the flight plan, and, in practice, single-date. A survey describes the canopy as it was on one summer afternoon. It says nothing about budburst, senescence, or the following year, and it does not extend one metre beyond its footprint.

1.4.2 Optical satellite imagery

Sentinel-2 occupies the opposite corner. The two-satellite constellation images every land surface every five days in thirteen spectral bands, four of them delivered at 10 m and six at 20 m. Three of the 20 m bands sit on the red edge, where they are sensitive to canopy chlorophyll and leaf area (Delegido et al., 2011; Drusch et al., 2012). The mission was designed for operational vegetation monitoring, and leaf area index is one of its standard biophysical products. The mixed resolution is not a detail for forest work. Restricting an inversion to the 10 m bands gains spatial detail and leaves the red-edge and shortwave infrared information unused, while including the 20 m bands does the reverse.

Table 1.2 lists the bands and the two native resolutions between which that choice is made.



every step is a modelling choice: the result is an estimate under stated assumptions, not a measurement

Figure 1.10. Schematic: how a LiDAR leaf area index is made, and where its assumptions enter. The gap fraction read off the normalised point cloud is inverted through the Beer-Lambert law into the plant area above each height, and differentiating that with height gives the density profile whose integral is the leaf area index. The extinction coefficient and the assumption of randomly placed foliage are annotated on the step that introduces them. No axis carries measured values.

Table 1.2. The ten Sentinel-2 bands used for vegetation, with their central wavelength and their ground sampling distance, as specified for the mission (Drusch et al., 2012). Three further bands, at 60 m, serve atmospheric correction and cloud screening rather than vegetation, and are not listed. The split matters for forest work because the red-edge and shortwave infrared bands, which carry the leaf area information least affected by saturation, are the ones acquired at 20 m. Chapter 2 restricts the inversion to the 10 m bands, on the grounds that spatial detail matters in heterogeneous canopy, and states the relevance of the 20 m bands as an open question.

Band	Centre wavelength (nm)	Native resolution (m)	Spectral region
B02	490	10	blue
B03	560	10	green
B04	665	10	red
B05	705	20	red edge
B06	740	20	red edge
B07	783	20	red edge
B08	842	10	near infrared
B8A	865	20	narrow near infrared
B11	1610	20	shortwave infrared
B12	2190	20	shortwave infrared

Retrieval proceeds by inverting a canopy radiative-transfer model against the observed reflectance. The standard model couples a leaf optical model with a canopy scattering scheme, giving PROSAIL (Feret & De Boissieu, 2024; Féret et al., 2017; Jacquemoud et al., 2009; Verhoef, 1984). Because inverting it directly is ill-posed, in the sense that different parameter combinations produce nearly identical spectra, the inversion is regularised by prior information on the parameter ranges (Combal et al., 2002). The neural network distributed with the Sentinel toolbox is trained on a simulated database built from prescribed parameter distributions (Weiss et al., 2020), and hybrid inversion trains a machine-learning regressor on a comparable look-up table (Verrelst et al., 2012; Verrelst et al., 2015; Verrelst et al., 2019). Both have been applied to forests (Chrysafis et al., 2020; Fernandes, Hong, et al., 2024) and validated against field data (Brown et al., 2021).

Turning that into a usable time series takes an operational chain. Top-of-atmosphere reflectance must be atmospherically corrected, clouds and their shadows masked, and observation and illumination geometry recorded, since the inversion depends on them (Feret, 2024). Cloud cover in temperate Europe removes a large share of the nominal five-day revisit, so the surviving observations are irregularly spaced. A continuous trajectory has to be reconstructed by gap-filling and smoothing before it can force anything (Atzberger & Eilers, 2011; Liang et al., 2023). The satellite therefore delivers a smoothed estimate of a seasonal trajectory rather than a set of measurements. That trajectory is its advantage over an airborne survey, because the timing of leaf-out and senescence is observed per pixel rather than assumed (Cotrina-Sanchez et al., 2026). It is not free of artifacts. Surface reflectance is directional, and the sun and view geometry of an acquisition change through the season independently of the canopy (Bréon et al., 2002; Maignan et al., 2004). A retrieved trajectory can therefore drift

after the summer solstice for reasons unrelated to leaf loss. Chapter 3 therefore treats the seasonality of Sentinel-2 as a signal to be tested rather than as a property to be assumed.

Four constraints qualify it, and together they define the gap set out in Section 1.5.2. The first is geometric: the signal is a two-dimensional reflectance measured from above, so it carries information about the upper canopy and only indirectly about what lies beneath. The second is the saturation that follows from it. Sensitivity declines as leaf area accumulates and is exhausted somewhere above five, the exact point depending on the retrieval, the index, and the canopy (Aklilu Tesfaye & Gessesse Awoke, 2021; Gao et al., 2023). Closed temperate forest canopies are at or beyond that limit for much of the growing season. The consequence is a compression rather than an abrupt ceiling: differences in leaf area between dense stands are progressively under-expressed in the reflectance before they disappear from it. The third is the prior information that regularises the inversion. Operational parameter ranges are tuned for crops, and transferring them to closed forest canopies biases the retrieval, while correcting the bias trades against variance (Fernandes, Djamai, et al., 2024). The fourth is spatial heterogeneity: because the reflectance-to-LAI relationship is non-linear, a pixel containing a mixture of canopy heights or densities does not return the average of its parts (Garrigues et al., 2006; Ma et al., 2008). Continuous temporal coverage therefore comes at the cost of structural depth, and the cost is largest exactly where the canopy is densest and most heterogeneous.

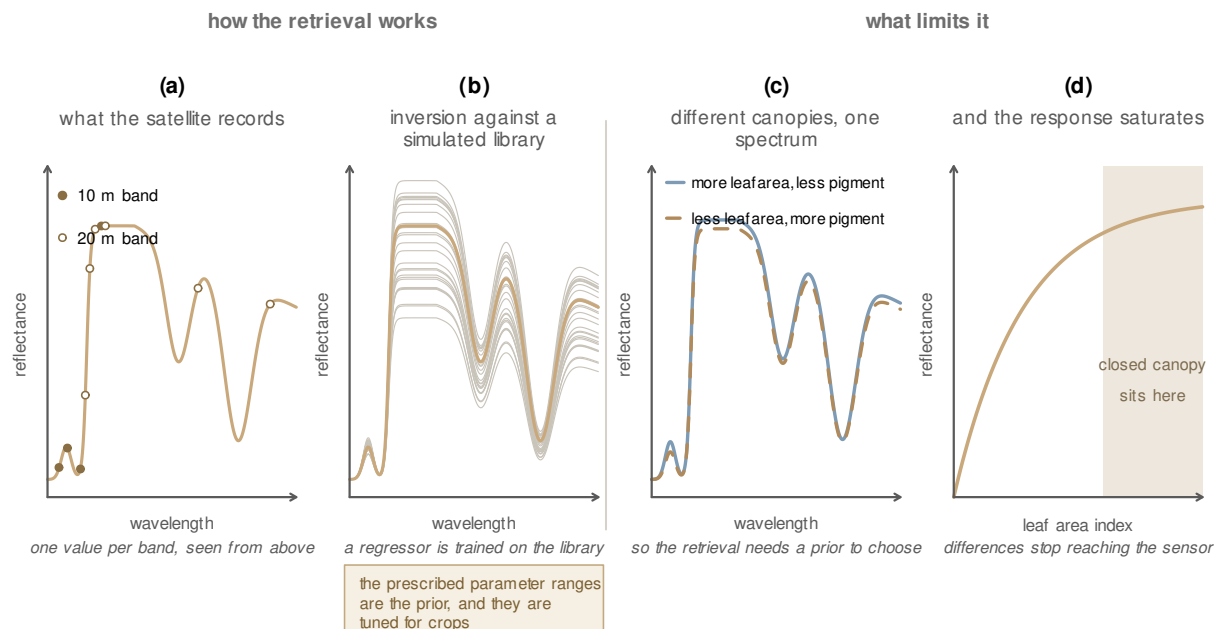
1.5 Identifying the scientific gap

1.5.1 The attribution gap

The first gap is one of causal attribution. Canopy structural variables covary strongly in real stands, so correlative microclimate models cannot assign the buffering effect to any one of them. Coefficients are unstable, smooth terms are concave, and machine-learning importance measures redistribute themselves among correlated predictors in ways that depend on the algorithm (Dominici, 2002; Gregorutti et al., 2017; Nicodemus et al., 2010). The question posed in Section 1.2.2, whether the quantity of foliage or its vertical arrangement governs sub-canopy buffering, therefore remains open despite an extensive literature on both sides (Ehbrecht et al., 2019; Kovács et al., 2017).

The gap is methodological. More field data collected along the same natural gradient will not close it, because the gradient itself confounds the predictors. Closing it requires observations in which the structural variables move independently of one another. No forest offers such observations, so they have to be generated by a mechanistic model driven by controlled combinations of canopy inputs. The perturbations are applied around each plot's own measured canopy, so every simulated configuration stays close to one that occurs in the field.

Simulation in place of observation raises one objection: a regression whose coefficients are unidentifiable has been traded for a model whose output cannot be checked. Three properties answer it. First, the model's parameters are physical and are not fitted to the response being explained, so the answer cannot enter through a calibration step.



wall-to-wall and repeated, but two-dimensional: the depth of the canopy is what the signal cannot reach

Figure 1.11. Schematic: how a Sentinel-2 leaf area index is made, and what limits it. The satellite records one reflectance value per band, seen from above; the retrieval trains a regressor on a library of spectra simulated over prescribed parameter ranges, and those ranges are the prior. Two limits follow. Different combinations of leaf area and pigment produce nearly the same spectrum, so the prior decides between them; and the response flattens once leaf area has accumulated, which is the regime closed temperate canopies occupy. No axis carries measured values.

The sensitivity of simulated temperature to leaf area follows from radiative and turbulent transfer. Second, the perturbations are small and centred on each plot's measured canopy, so the physics is never extrapolated into regimes where it has not been evaluated. Third, the model's ability to reproduce the observed between-plot buffering is established separately against the logger network. The attribution is therefore made with a model whose behaviour on real stands has been characterised independently of it. No design gives a direct empirical test of the attribution itself, but the physics it rests on can be tested independently of the gradient at issue, unlike a fitted coefficient.

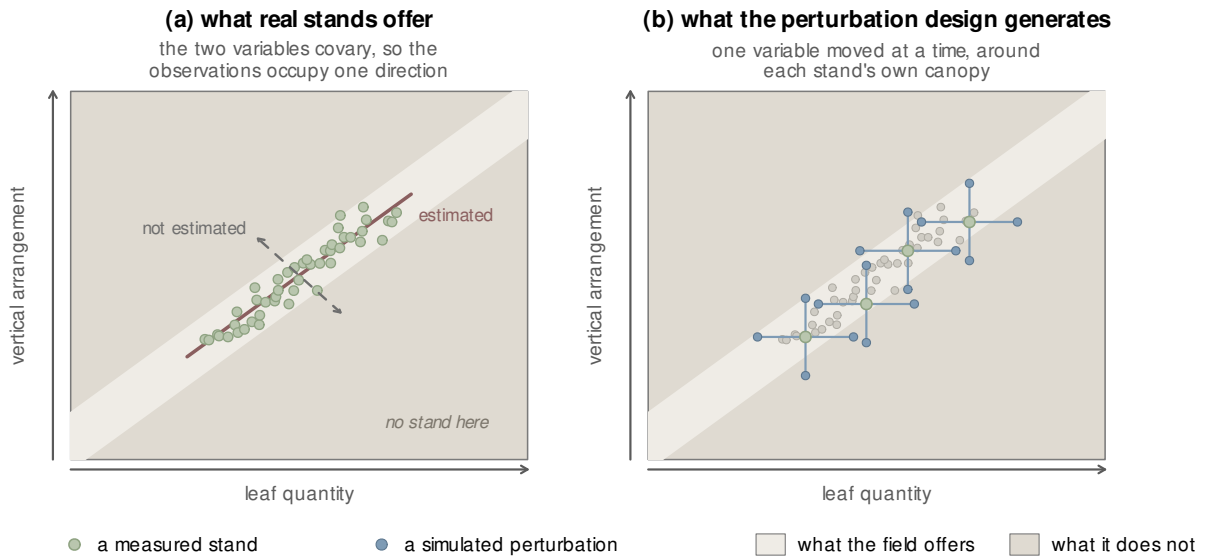


Figure 1.12. Schematic: why the attribution question cannot be settled on observations alone. In real stands the structural variables covary, so the observations occupy a narrow band of the space they span; a regression fitted to that band estimates the direction along it and leaves the direction across it unidentified, and further sampling along the same gradient adds no information about the second direction. Moving one variable at a time around each stand's own measured canopy generates the combinations the field does not offer. No axis carries measured values.

1.5.2 Optical versus structural LAI

The second gap concerns the measurement of leaf area itself. Airborne LiDAR and Sentinel-2 both deliver a quantity called LAI, but they measure different physical things. LiDAR integrates intercepted returns through the whole canopy column. The optical retrieval inverts a reflectance signal that originates mostly near the top (Gara et al., 2019) and stops responding once the canopy is dense (Baret & Guyot, 1991; Gao et al., 2023). The two are known to disagree, and the disagreement is expected to grow with canopy density, but its magnitude and its drivers have not been characterised jointly for temperate deciduous forest.

Three candidates contribute, and they are rarely assessed together. Optical attenuation limits the canopy depth that contributes to satellite reflectance, so the satellite may describe only the upper part of the column that the LiDAR integrates in full. The radiative-transfer inversion supplies prior ranges tuned for crops rather than for forest canopies, and those priors propagate into the retrieved leaf area (Combal et al., 2002; Fernandes, Djamai, et al., 2024). Horizontal heterogeneity within a satellite pixel, which combines variation in canopy height with variation in the terrain below, strains the one-dimensional assumption the inversion rests on, and does so non-linearly (Garrigues et al., 2006).

The question is one of inter-sensor consistency, a prerequisite rather than a microclimate claim. No ground-based LAI measurement coincident with the airborne and satellite acquisitions exists at these sites, so nothing external can decide between the two retrievals and the target cannot be absolute accuracy. The reachable target is which part of the canopy the satellite number describes, and it has to be settled before a satellite LAI forces a microclimate model.

Two caveats condition everything downstream. The first is domain. Consistency between two sensors holds over the structural domain on which it was established. A depth-based comparison has to restrict itself to canopy deep and closed enough to have an interior at all. A search over successive canopy depths cannot converge on stands shallower than the depths it scans, so pixels below a minimum canopy height are excluded. A high fractional-cover threshold keeps gaps and edges out of the pixel. Any agreement reached on that restricted domain is silent about short, open stands, where an optical retrieval behaves differently.

The second caveat is that better agreement in leaf area does not propagate into better simulated microclimate. A mechanistic canopy model responds non-linearly to leaf area, steeply in open canopy and with strong saturation once the canopy closes. The same reduction in leaf-area error is therefore worth different amounts depending on where along that curve it applies. A retrieval tuned to minimise leaf-area discrepancy over one part of the range can be a worse driver of the microclimate over another. It can even reverse the ordering of plots, if the tuning compresses the gradient the model relies on. Which of the two objectives a correction improves, leaf-area accuracy or microclimate skill, is an empirical question.

1.5.3 The spatio-temporal trade-off

The third gap follows from the first two. ALS resolves three-dimensional structure but is acquired once and over a limited footprint. Sentinel-2 is repeated and spatially complete, but optical, two-dimensional and saturating. Spaceborne LiDAR samples the vertical profile repeatedly, in sparse footprints (Dubayah et al., 2020). No single sensor currently delivers fine three-dimensional structure and temporal dynamics together.

Driving a mechanistic model beyond the plots where LiDAR happens to exist therefore forces a compromise between structural fidelity and coverage. That compromise has usually been made implicitly, by using whichever product was available, and evaluated by pooling all plots into a single accuracy statistic. Neither the choice nor the evaluation has been examined against the regime dependence that Sections 1.4.2 and 1.5.2 predict. If the optical signal fails specifically where the canopy is dense, a single

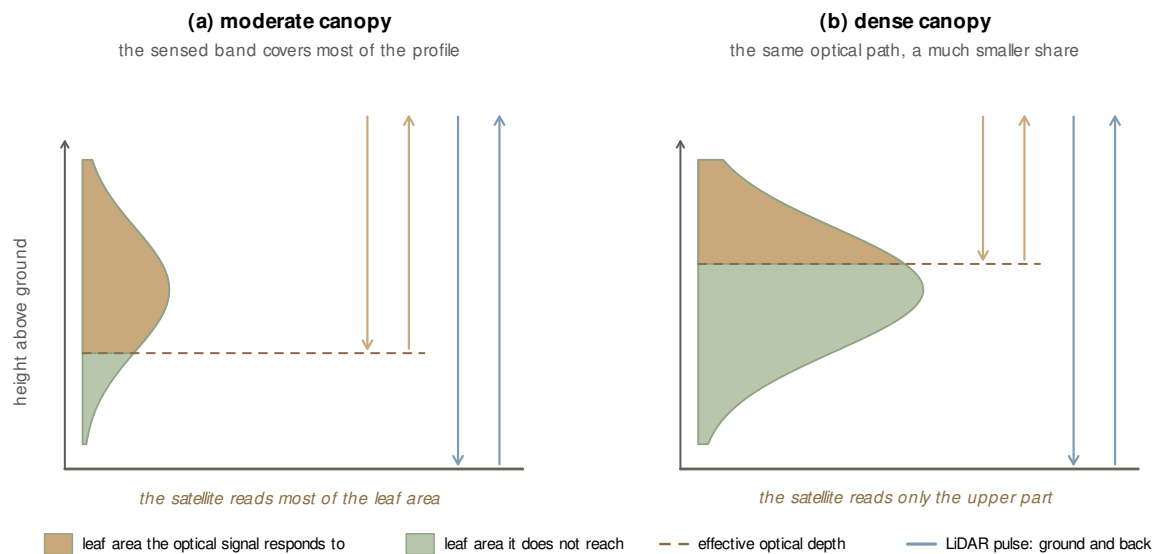


Figure 1.13. Schematic: the two sensors do not read the same part of the canopy column. A laser pulse travels to the ground and back, so gap-fraction inversion integrates the whole profile. Reflected sunlight stops carrying information once enough leaf area lies above, so an optical retrieval responds only to the leaf area shallower than an effective optical depth. Holding that optical path fixed while leaf area increases, from (a) to (b), leaves the satellite reading a steadily smaller share of the column. The depth drawn here is a fixed optical path, so it sits higher in the denser canopy; Chapters 2 and 3 approximate it as a fixed geometric depth below the canopy top. This is the structural origin of optical saturation; no saturation curve is plotted, and no axis carries measured values.

pooled conclusion on which sensor is better can conceal two opposite conclusions in the two halves of the density gradient. Which compromise is right, and whether it is the same compromise everywhere in a landscape, has not been established.

The three gaps have to be addressed in order. The structural dimension the microclimate responds to defines what a sensor must measure. Whether Sentinel-2 measures it depends on the part of the canopy the optical signal describes. The choice of forcing product can only be judged once both are known, and the order of the chapters follows the order of the gaps.

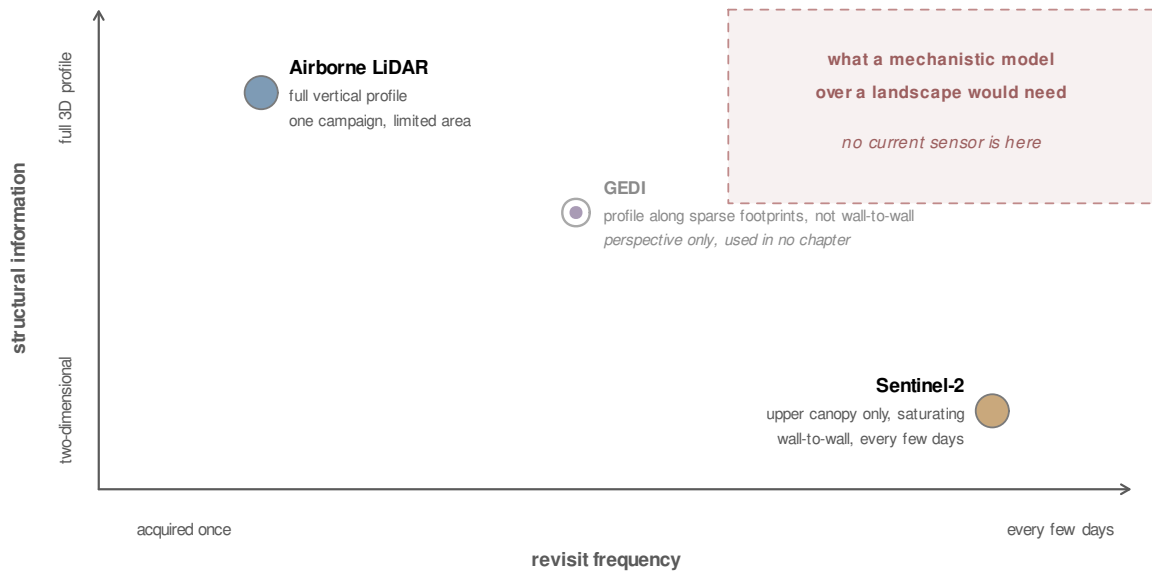


Figure 1.14. Schematic: the spatio-temporal trade-off between structural sensors. Each sensor is placed by how much of the three-dimensional canopy it recovers and how often it revisits; spatial coverage, a third property, is stated in words rather than encoded in symbol size. The corner a mechanistic microclimate model would need, full structure repeated through the season, is empty, and that emptiness is the third gap. Airborne LiDAR and Sentinel-2 are the two sensors this thesis works with. GEDI is drawn hollow: it is a perspective taken up in the General Discussion and is used in none of the three chapters, and it is shown here only so that the empty corner is not left open to the objection that spaceborne LiDAR already fills it. Positions are qualitative and no axis carries tick values.

1.6 Research questions and objectives

RQ1 (Chapter 1). *Which dimension of canopy structure governs sub-canopy ΔT_{max} buffering, leaf quantity or vertical arrangement, and can a mechanistic model attribute it despite the collinearity of structural variables?*

The objective is to couple ALS-derived canopy structure (LAI, fCover, H_{max} , and the LAD profile) to MuSICA and to run a controlled trait-perturbation sensitivity analy-

sis. Each structural variable is moved by a small, native step around every plot's own canopy, and the measured vertical profile is contrasted against a leaf-area- and height-preserving uniform profile. Perturbing each plot around its own operating point gives the local response of a real canopy rather than an extrapolation. The resulting effects are stratified by structural archetype, so the answer may differ between open and dense canopy instead of being averaged into one conclusion. The simulated buffering is validated against an in-situ network of 53 temperature loggers. Two hypotheses are posed. The first is that vertical structure modulates radiative and turbulent transfer enough for the LAD profile to change simulated $\Delta T_{\max}$ measurably. The second is that, at fixed leaf area and height, foliage concentrated high in the canopy attenuates the diurnal cycle more strongly than foliage spread through the column.

RQ2 (Chapter 2). *Can Sentinel-2 optical LAI be reconciled with LiDAR-derived structural LAI, or does optical saturation decouple them?*

The objective is to intercompare the two retrievals across three temperate deciduous forests and to assess the three candidate drivers of their discrepancy identified in Section 1.5.2. These are the effective canopy depth contributing to the satellite reflectance, the parameterisation of the PROSAIL inversion, and horizontal heterogeneity of the canopy surface within the satellite pixel. Each correction is applied on top of the previous one, so their individual contributions can be read. The target is inter-sensor consistency, not absolute accuracy, since no coincident field measurement of LAI is available to decide between the two sensors. The output is a quantified statement of how well, and under what canopy conditions, the two retrievals can be brought into agreement, to be carried forward as a condition of use.

RQ3 (Chapter 3). *Which remotely sensed leaf area should force a mechanistic microclimate model, does the answer depend on canopy archetype, and can the seasonality of Sentinel-2 track the microclimate without losing the accuracy of LiDAR?*

The objective is to force MuSICA with four leaf-area scenarios, holding canopy height and the shape of the vertical profile at their LiDAR values, so that the design isolates the effect of the leaf-area source. The first scenario holds the LiDAR leaf area fixed. The second and third use the seasonal Sentinel-2 retrieval, first as delivered and then forest-tuned. The fourth applies the Sentinel-2 seasonality to the LiDAR magnitude.

One contrast organises the chapter: a satellite leaf area that describes the upper canopy against a LiDAR leaf area that integrates the whole column. The deficit expected in dense canopy is the leaf area lying below the depth the optical signal reaches. The archetypes of Chapter 1 separate the canopies where that deficit matters from those where it does not. The scenarios are scored against the logger network, archetype by archetype, on a within-plot paired sensitivity test and a direct validation, with the coupling slope as primary metric and the daytime offset as secondary. We hypothesise that no single product forces the microclimate best in every archetype: seasonal Sentinel-2 should rank the plots where it does not saturate, and where it saturates its magnitude should come from LiDAR.

1.7 Overview of the thesis

The three chapters share one question but not one design. Table 1.3 sets out what each one uses, and Figure 1.16 locates the forests they work in. Each chapter narrows

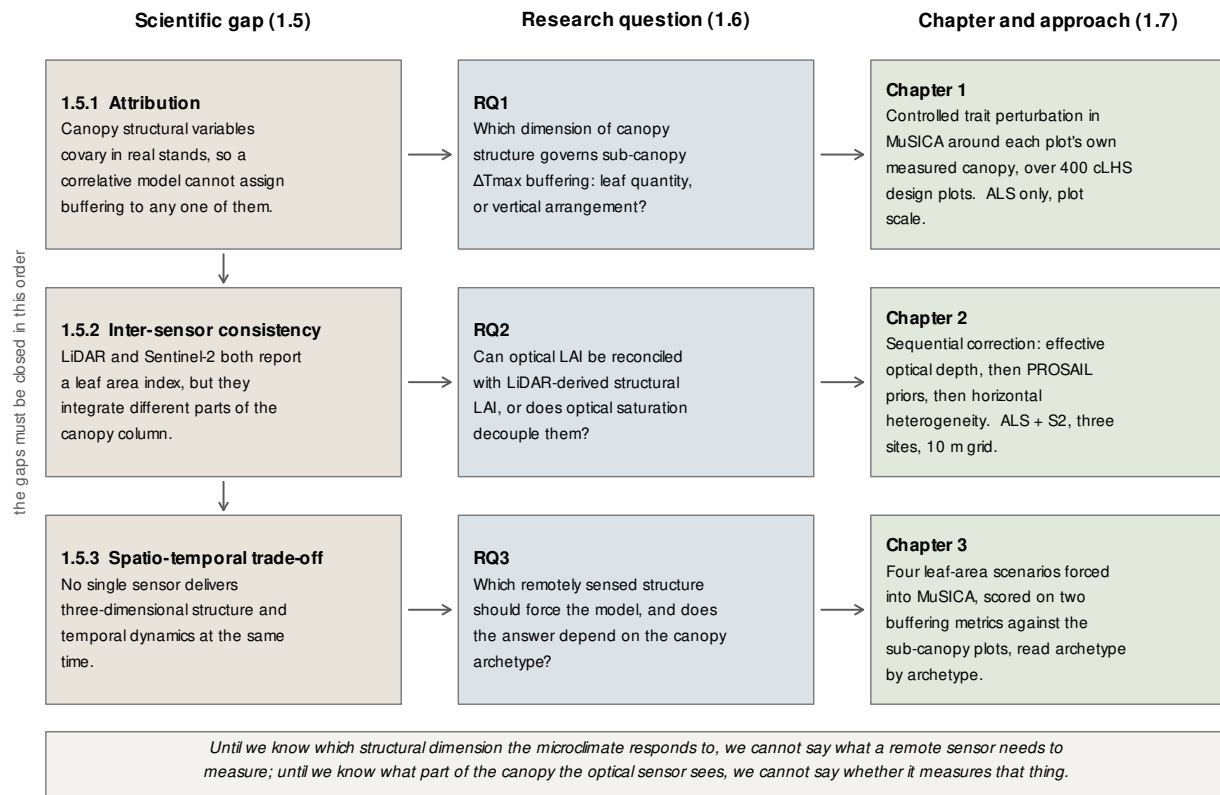


Figure 1.15. The structure of the thesis. Each of the three gaps identified in Section 1.5 gives rise to one research question and one research chapter. The vertical arrows on the left indicate that the gaps are not independent: each has to be closed before the next can be posed, which sets the chapter order.

the question the next one asks.

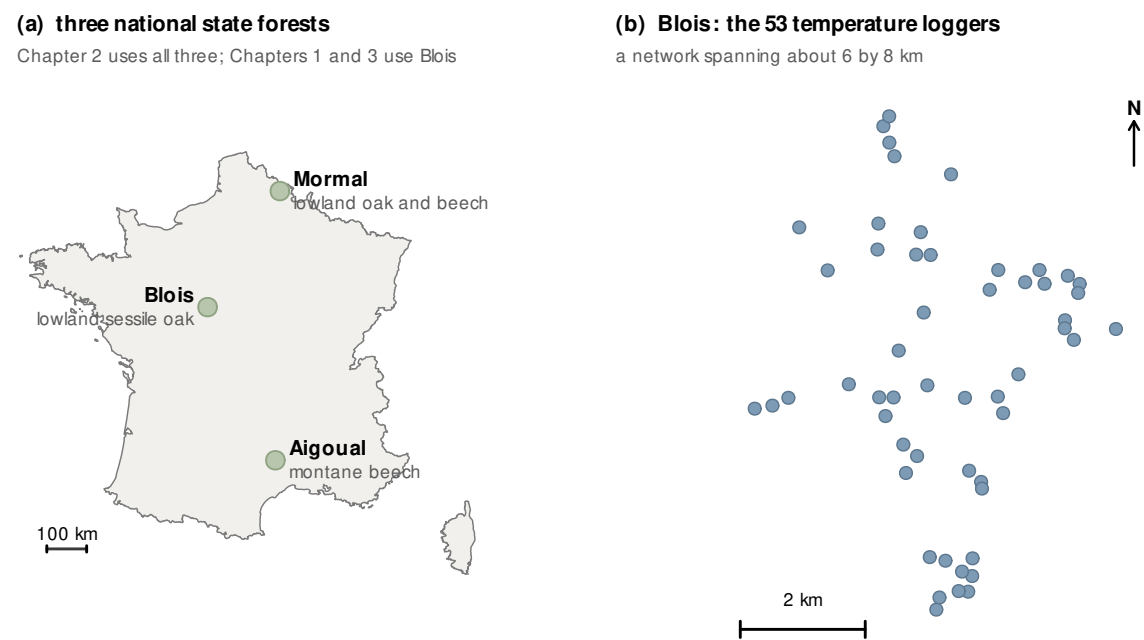


Figure 1.16. The study system. **(a)** The three national state forests, all managed by the French National Forests Office and drawn from a national forest monitoring network: Aigoual, a montane beech forest in the south; Blois, a lowland forest almost entirely of sessile oak; and Mormal, a lowland oak and beech forest in the north. Chapter 2 uses all three, Chapters 1 and 3 use Blois alone. **(b)** The 53 temperature loggers at Blois, against which every simulated offset in Chapters 1 and 3 is validated. They are drawn in one colour because their structural typology is a result of Chapter 1, not a property of the network. Unlike the other figures of this chapter, this one carries measured positions.

Table 1.3. What each research chapter uses. The progression is one of constraint rather than of scale. Chapter 1 establishes which structural variable the microclimate responds to. Chapter 2 asks whether a satellite can measure that variable. Chapter 3 asks which retrieval should then drive the model. Only Chapter 2 leaves the single site, and it does so to test whether the optical discrepancy is general rather than local.

	Chapter 1	Chapter 2	Chapter 3
Question	Which dimension of structure governs buffering, and can it be attributed?	Can optical and structural leaf area be reconciled?	Which retrieved leaf area should force the model, and in which canopy regime?
Forest	Blois	Aigoual, Blois, Mormal	Blois
Sensors	Airborne LiDAR	Airborne LiDAR and Sentinel-2	Airborne LiDAR and Sentinel-2

	Chapter 1	Chapter 2	Chapter 3
Mechanistic model	MuSICA	none	MuSICA
Working resolution	20 m	10 m	10 m
Unit of analysis	400 design plots, each perturbed around its own canopy	pixels across three massifs	53 logger plots, under four leaf-area scenarios
Judged against	the perturbation design itself, no field measurement entering the chapter	inter-sensor consistency, no field leaf area being available	53 temperature loggers, archetype by archetype

Chapter 1 answers RQ1. ALS-derived structure from a lowland sessile oak forest drives MuSICA at 400 design plots, stratified into four structural archetypes from open to dense. Each structural variable is perturbed around every plot's own canopy, so its contribution to simulated $\Delta T_{\max}$ is isolated by construction rather than inferred from correlated observations. The answer depends on canopy density, which reconciles the conflicting literature of Section 1.2.2. Leaf area and cover together dominate, and both steepen as the canopy closes. In the most open stands every effect falls below 0.03 °C, too small to rank. The vertical profile is second order. Outside those open stands the real-versus-uniform contrast stays at or below the cover step. Its rank against the smaller leaf-area step depends on the response metric and the summary statistic. No field measurement enters Chapter 1: the perturbation design points are not the logger plots, and the coupled model is evaluated against the 53 loggers in Chapter 3. Because the microclimate responds to leaf quantity, the next question is whether a satellite can measure it.

Chapter 2 answers RQ2. Sentinel-2 and LiDAR LAI are intercompared across three temperate deciduous forests, a montane beech stand and two lowland oak-dominated ones, and the three candidate drivers of their discrepancy are addressed in sequence. Satellite LAI agrees best with the LiDAR estimate when the latter is restricted to the upper 6 to 10 m of the canopy. That depth is empirical, the one at which the two sensors are most consistent, not a measured optical penetration. Adapting the PROSAIL parameter ranges to forest canopies then improves agreement unevenly, and most of that gain survives when the LiDAR-derived priors are replaced by LiDAR-free ones. Residual discrepancies remain largest where the canopy surface is horizontally heterogeneous within the pixel, so vertical thickness and horizontal heterogeneity have to be accounted for jointly. What improves is inter-sensor consistency, not accuracy against a field truth, and it holds only on the closed, tall canopy the comparison was posed on. Chapter 3 reads this as a statement of where the optical signal can be trusted, not as a correction to apply everywhere.

Chapter 3 answers RQ3. MuSICA is forced at the 53 logger plots, not over a landscape, with four leaf-area scenarios, and the best product depends on the canopy archetype. In the two densest archetypes Sentinel-2 ranks the plots measurably worse than LiDAR, and in the more open ones the 8 and 12 plots cannot separate them. In dense

canopy Sentinel-2 saturates, and the ranking is held only by products carrying the LiDAR magnitude. In the open gaps Sentinel-2 retrieves a leaf area that the LiDAR point cloud shows is not there. The mechanism is the one diagnosed in Chapter 2. Most of the leaf area of a dense canopy lies below the effective optical depth, so the Sentinel-2 deficit grows with the fraction buried below it. Section 1.5.2 anticipated the consequence. The forest-tuned retrieval that is most consistent with LiDAR drives the dense microclimate worse than the raw one, because a tuning that reduces the mean discrepancy flattens the between-plot contrast the model needs. The summer window is a weak test of the seasonal shape, which is almost flat on a full-leaf plateau, so what the combination adds can only be decided at leaf-out and senescence.

This thesis does not produce a wall-to-wall microclimate map. It establishes what a map would have to be built on. That is which structural dimension the temperature responds to, what a satellite can see of it, and which product should drive the model in which archetype. Mapping stays out of reach for two reasons established in the chapters. The forcing still requires LiDAR structure wherever the canopy is dense, so coverage is bounded by the flight footprint. And a one-dimensional column cannot represent the gaps, edges, and lateral exchanges a landscape is made of.

The microclimate work rests on a single lowland oak site and on the summer 2021 window, so the archetype-dependent conclusion of Chapter 3 will require testing elsewhere. The loggers are not spatially independent, which makes its confidence intervals optimistic. Within summer a temporally resolved satellite series adds nothing, because leaf area is on its phenological plateau and there is no timing left to resolve. Outside summer the chapters do not test the question, and the General Discussion draws on the later logger archive for perspectives only. On the optical side no ground LAI coincident with the acquisitions is available, so Chapter 2 establishes consistency rather than accuracy.

Two avenues are left open and treated in the General Discussion. The first is spaceborne LiDAR. GEDI and the global canopy-height products derived from it promise three-dimensional structure with satellite coverage (Dubayah et al., 2020; Lang et al., 2022; Schwartz et al., 2023). Plant area index products are being validated from the same instrument (Brown, Morris, Meier, et al., 2023), and fusion frameworks with Sentinel-2 are emerging for wall-to-wall retrieval (Jia et al., 2026). Its sampling geometry and geolocation uncertainty remain too coarse for plot-scale mechanistic microclimate modelling (Schleich et al., 2023).

The second avenue is dimensional. Forest edges are warmer than interiors (Reek et al., 2025), and the residuals of Chapter 1 are structured by the sub-pixel gaps a one-dimensional column cannot see. Heterogeneity- and edge-aware formulations that couple a canopy scheme to a three-dimensional atmospheric solver relax that assumption [REF: Grulois 2026, coupled ARPS-MuSICA thesis; Van De Walle et al. (2026)]. The archetype-dependent forcing of Chapter 3 can then be evaluated where lateral fluxes matter.

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